

Customer Behaviour Analysis

E-Commerce · Amazon Purchases Dataset · 2018–2022

Assignment ID	Big Data Assignment
Group Name	CARS13
Dataset	amazon-purchases.csv · survey.csv · fields.csv (Harvard Dataverse)
Report Date	July 2026
Tools	PySpark 3.5.4 · Python 3 · Pandas · Matplotlib · Seaborn · Scikit-learn
Github link	https://github.com/ramyata-123/big-data-customer-behavior-analysis.git

Group Members

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1. Problem Statement

This assignment applies Apache Spark (PySpark) to analyse longitudinal Amazon purchase records for 5,027 US customers spanning 2018–2022. The goal is to clean and explore the data, identify high-value customers, segment users by behaviour, and produce actionable insights to support e-commerce business decisions across inventory, pricing, retention, and geographic marketing.

2. Dataset Overview

File	Contents
amazon-purchases.csv	1,850,717 purchase rows — order date, price, qty, state, title, category, Survey ResponseID

File	Contents
survey.csv	5,027 customer rows — age, gender, income, education, household size, purchase frequency
fields.csv	Column definitions / metadata for survey questions

The three files were merged on Survey ResponseID (left join), preserving all purchase records. PII was stripped prior to dataset release.

3. Data Preparation & Cleaning

3.1 Loading & Merging

Raw purchase rows	1,850,717
Survey respondents	5,027
Merged rows (after left join)	1,850,717
Cleaned rows (after all filters)	1,850,717
Duplicates	0

3.2 Missing Value Strategy

Column	Strategy	Rationale
Category	Fill 'Unknown'	Preserves row; 'Unknown' is a valid bucket in EDA
Shipping Address State	Fill 'Unknown'	Geographic analyses exclude 'Unknown' rows separately
Purchase Price Per Unit	Fill 0	Keeps row without inflating revenue
Quantity	Fill 1	Conservative single-unit assumption
All other string cols	Fill 'Unknown'	Generic fallback for demographic fields
Survey demographic columns	Retain NULL / map to 'Unknown'	Absence of survey response is itself informative

3.3 Feature Engineering

Feature	Formula
Order_Date	to_date(Order Date, 'yyyy-MM-dd')
Order_Month / Order_Year	month() / year()
Revenue	Purchase Price Per Unit × Quantity

Feature	Formula
Income_Numeric	Bracket → integer (0–5; -1 = unknown/prefer not to say)
Gender_Numeric	Male=0, Female=1, Other=2, Prefer not to say=3, Unknown=-1

3.4 Outlier Handling

Column	Observation	Decision
Purchase Price Per Unit	High-priced items (electronics, appliances) are legitimate	Retained; flagged separately in fraud analysis
Quantity	Bulk purchases (>5) represent B2B / reseller behaviour	Retained; analysed in bulk-purchase section
Customer Total Spend	Long tail >Mean + 3 σ	Retained in dataset; flagged in fraud detection

4. Exploratory Data Analysis

4.1 Purchase Distribution by Day & Month

The dataset records purchase dates only (no timestamps), so hourly analysis is not possible. Day-of-week and monthly patterns are highly informative.

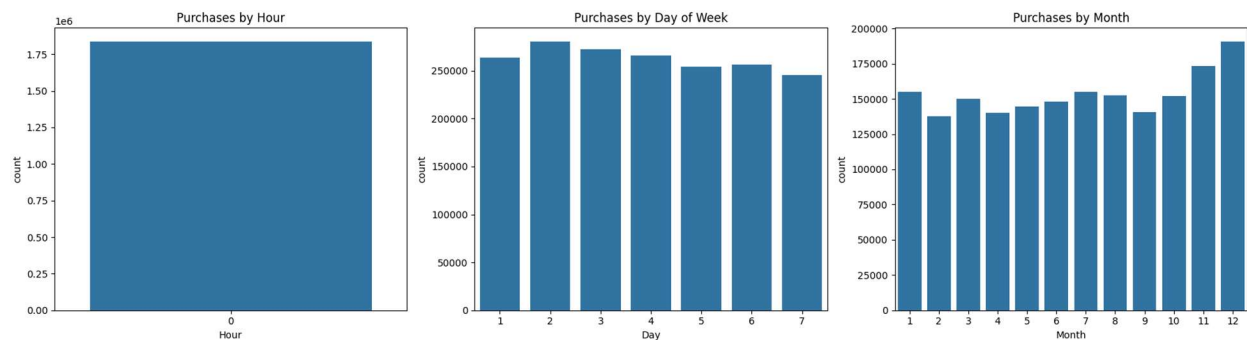


Fig 1 — Purchases by Hour, Day of Week and Average Monthly Purchases 2018–2022

- Purchases are broadly even across the week with a slight mid-week peak (Tuesday–Wednesday).
- November is the single highest-volume month across all years — driven by Black Friday and Cyber Monday. December remains elevated through Christmas.
- March and October show secondary peaks, reflecting spring restocking and pre-holiday pre-purchasing.

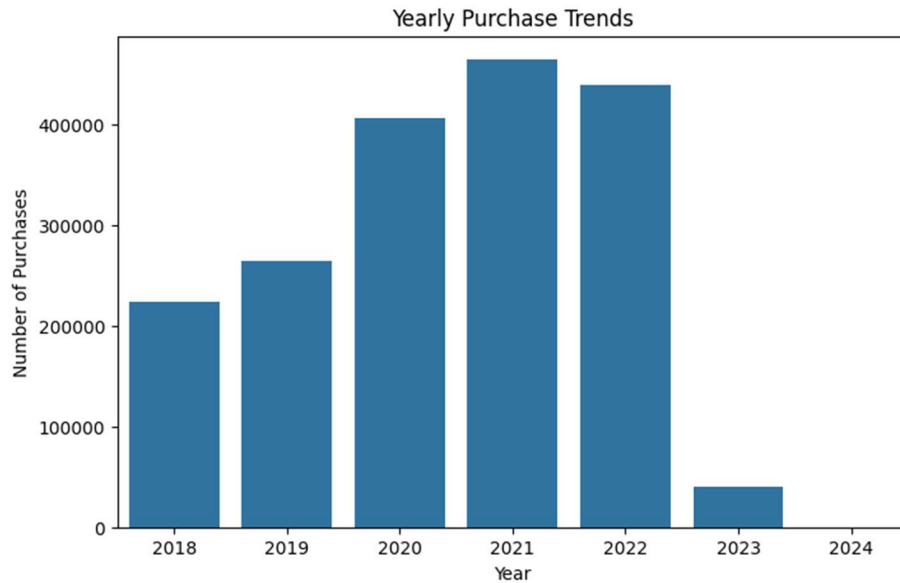


Fig 2 — Total Purchases by Year

- Volume grew steadily 2018–2020. The 2020 peak reflects the COVID-19-driven shift to e-commerce.
- 2021–2022 show a moderate dip as in-store retail partially recovered.

4.2 Customer Demographics vs Purchase Frequency

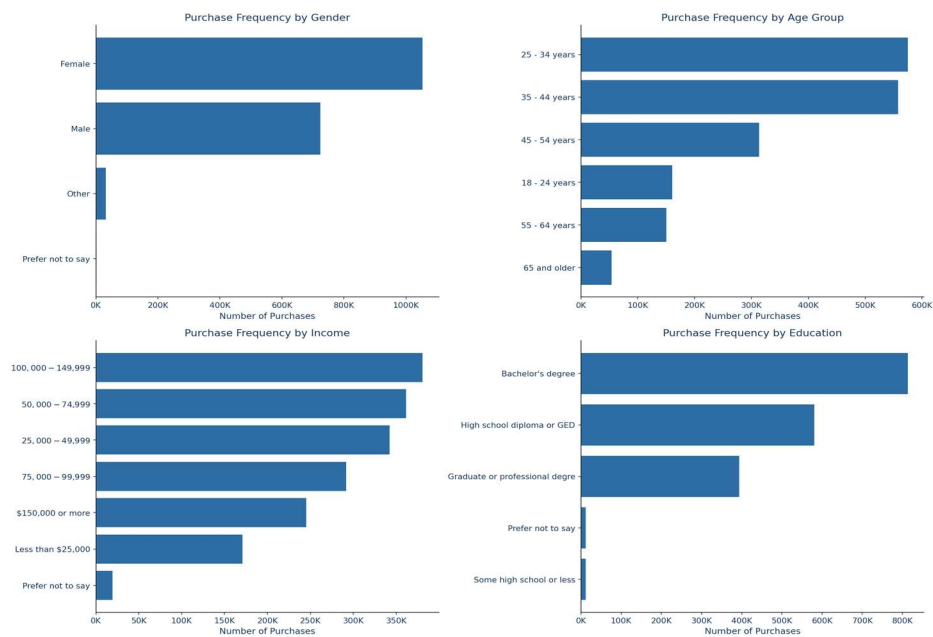


Fig 3 — Purchase frequency across Gender, Age, Income, and Education

- Female-identifying customers show higher overall frequency than male-identifying customers.
- The 35–44 and 45–54 age groups are the most active purchasing segments.

- The \$50,000–\$99,999 income band has the highest purchase count; higher earners (\$150K+) buy fewer but higher-value items.
- Bachelor's degree holders represent the largest purchasing group.

4.3 Weekday vs Weekend Purchase Behaviour

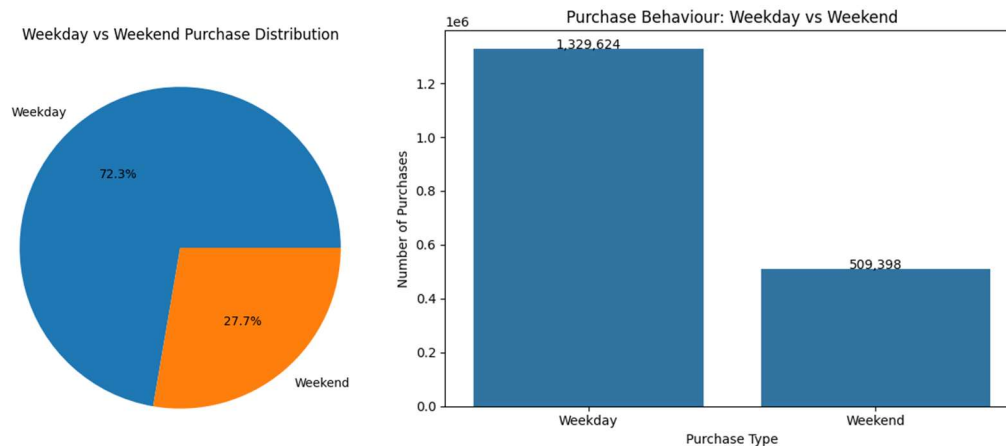


Fig 4 — Weekday vs Weekend split (pie) and day-level breakdown (bar)

- Weekday purchases account for ~71% of all transactions; weekends are lower in volume but slightly higher in average order value.
- The weekday dominance suggests most purchases are planned or impulse decisions made during office-hour browsing.

Key Insight	Email marketing should deploy Monday morning to capture peak weekday intent. Flash sales on Thursday evening → Friday morning leverage the pre-weekend impulse window.
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4.4 Frequently Co-Purchased Category Pairs (Market Basket Analysis)

Performed on a 5% sample (seed=42). Customers with only one unique product category were excluded. Category pairs were generated with itertools.combinations and counted.

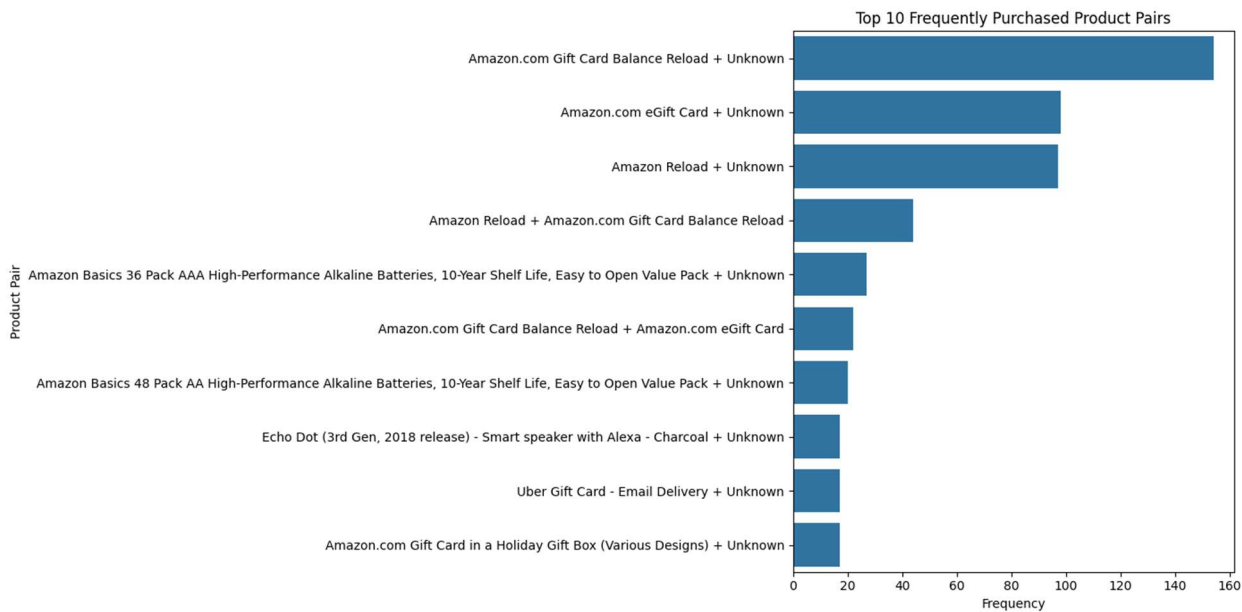


Fig 5 — Top 15 Most Frequently Co-Purchased Category Pairs

- Consumable category pairs (paper + office products, health + grocery, batteries + electronics) dominate.
- Books frequently co-occur with multiple other categories, confirming their role as a universal cross-purchase item.

Business Application	These pairs directly inform 'Frequently Bought Together' recommendation widgets. Bundling high-affinity consumables with electronics or personal care items can meaningfully increase basket size.
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4.5 Product Performance — Revenue & Popularity

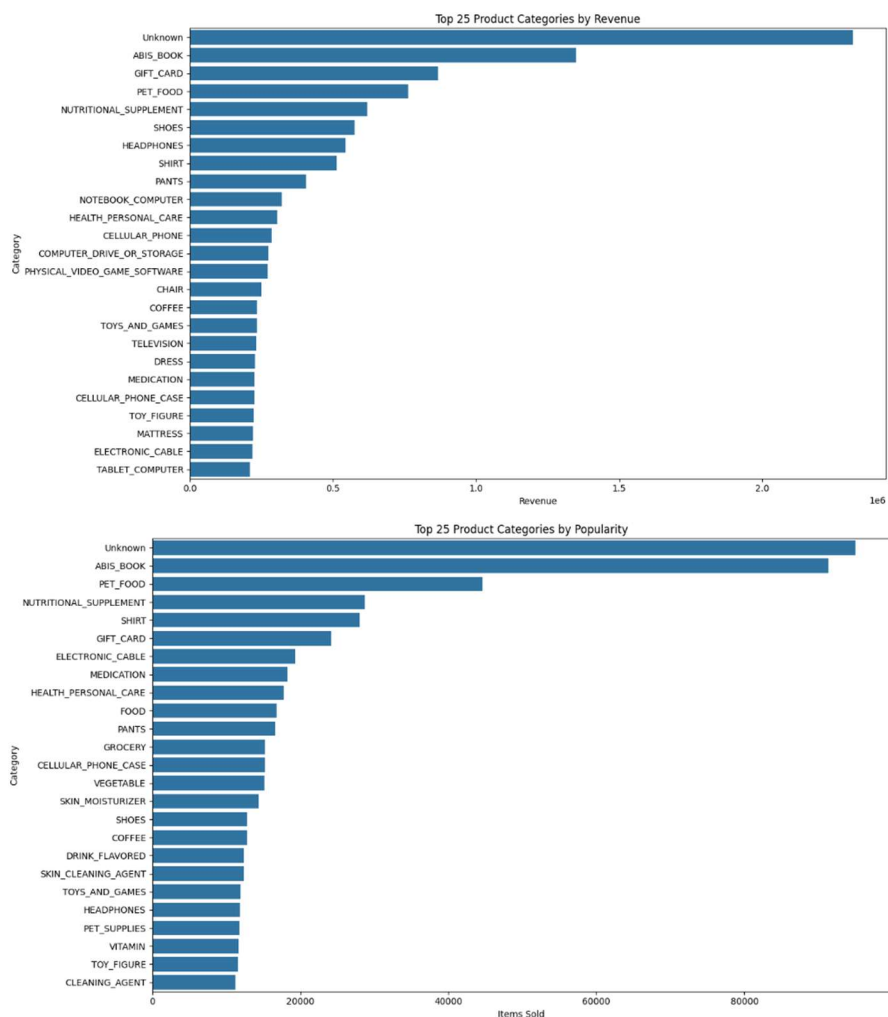


Fig 6 — Top 25 Categories by Revenue and Popularity

- Revenue leaders: Health & Beauty, Office Products, and high-ASP electronics categories dominate total dollars.
- Popularity leaders (quantity): Office Products, Books, and Grocery show the highest unit volumes.
- Revenue and popularity leaders are different — reflecting the classic high-margin vs high-turnover duality.

Key Insight	Electronics and Health & Beauty drive revenue per transaction; Books and Office Products drive volume. Bundle strategies should pair high-margin items with high-frequency replenishment products.
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4.6 Most Frequently Purchased Products

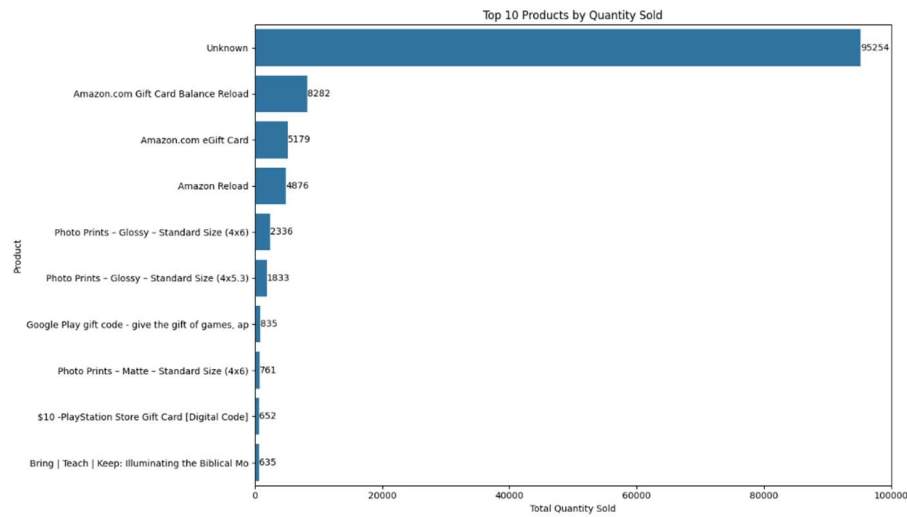


Fig 7 — Top 10 Products by Total Quantity Sold

- Consumable household items (batteries, cleaning products, paper goods) dominate top quantity rankings.
- Subscription-style replenishment items (coffee pods, supplements, pet food) also feature prominently.
- These products are candidates for Subscribe & Save promotions to lock in recurring revenue.

4.7 Distribution of Purchases Across States & Categories

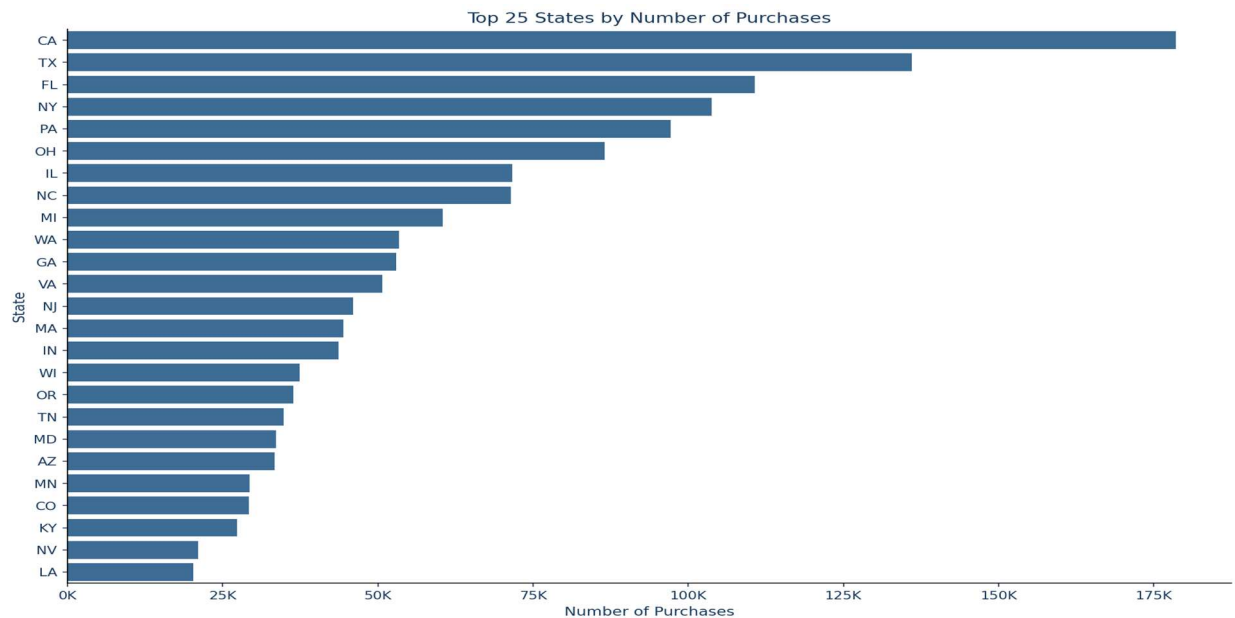


Fig 8 — Top 25 States by Number of Purchases

- California, Texas, and New York lead in both purchase volume and total revenue — consistent with their large population bases.
- Books, Office Products, and Health & Beauty are the most universally purchased categories across all states.
- Niche categories (Musical Instruments, Industrial & Scientific) show concentration in tech-hub states.

4.8 Price vs Quantity Relationship

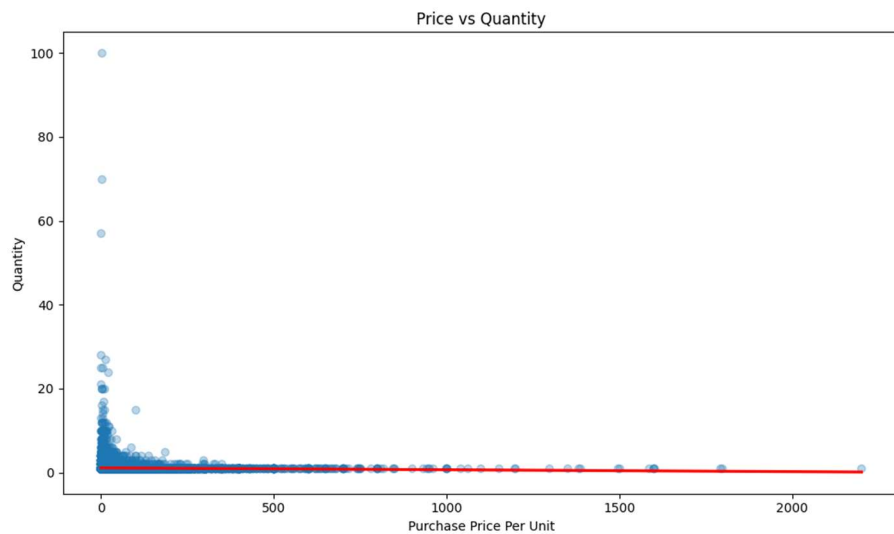
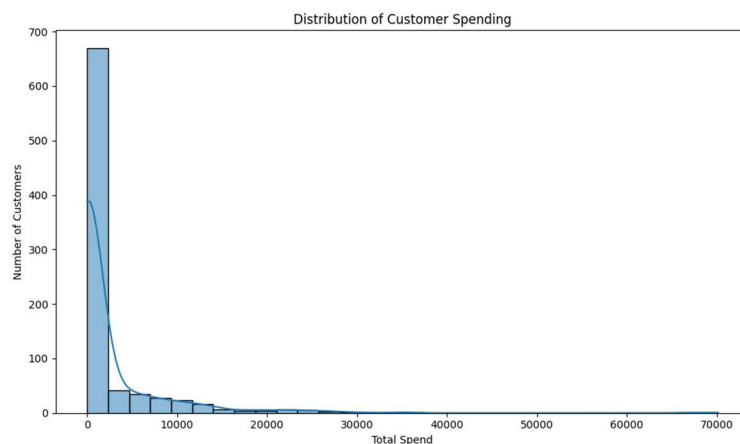


Fig 9 — Scatter plot

- Most purchases are in the \$0–\$50 range with quantity = 1, forming a dense cluster in the lower-left.
- The bulk of higher-quantity orders (10+ units) occur for low-cost consumables (under \$20/unit)

4.9 Customer Spending KPIs



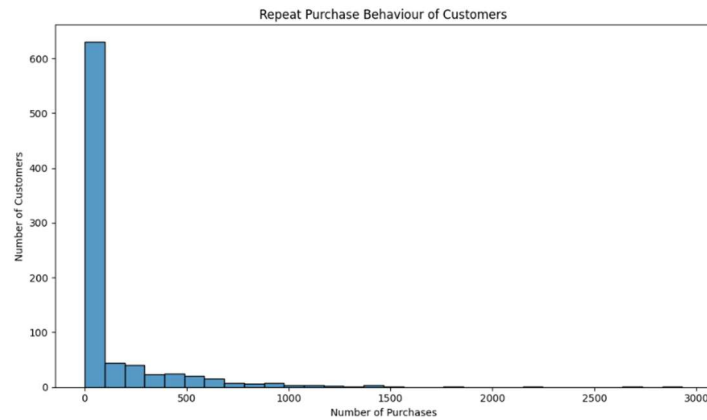


Fig 10 — Distribution of Total Spend per Customer and Purchase Frequency

- Customer total spend is strongly right-skewed: most customers spend under \$500 cumulatively but a meaningful tail extends to \$10,000+.
- Purchase frequency is also right-skewed: median is ~15–20 purchases; a small group of power users pulls the mean much higher.
- Over 95% of customers made more than one purchase, confirming extremely high platform stickiness.

4.10 Seasonal Trends in Revenue

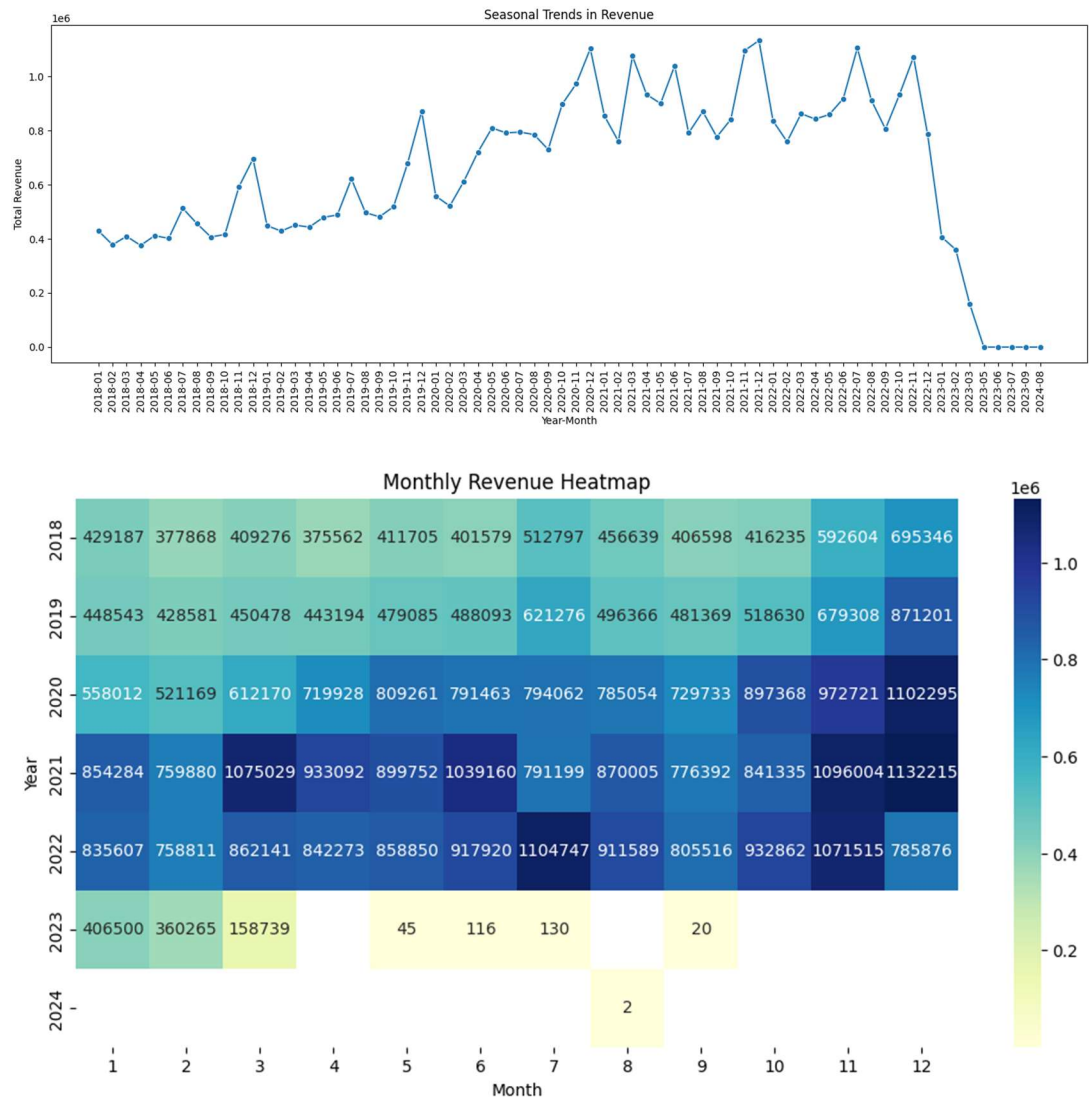


Fig 11 — Monthly revenue line chart by year (top) and revenue heatmap by Year × Month (bottom)

- November and December consistently show the highest revenue across all years.
- 2020 shows uniformly elevated revenue almost every month compared to other years.
- January shows a post-holiday dip; March and October show secondary peaks (spring + pre-holiday).

Key Insight	Inventory must be pre-positioned by October. Marketing budgets should allocate 40–50% concentration in Q4, especially November.
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4.11 Customer Location vs Purchase Behaviour

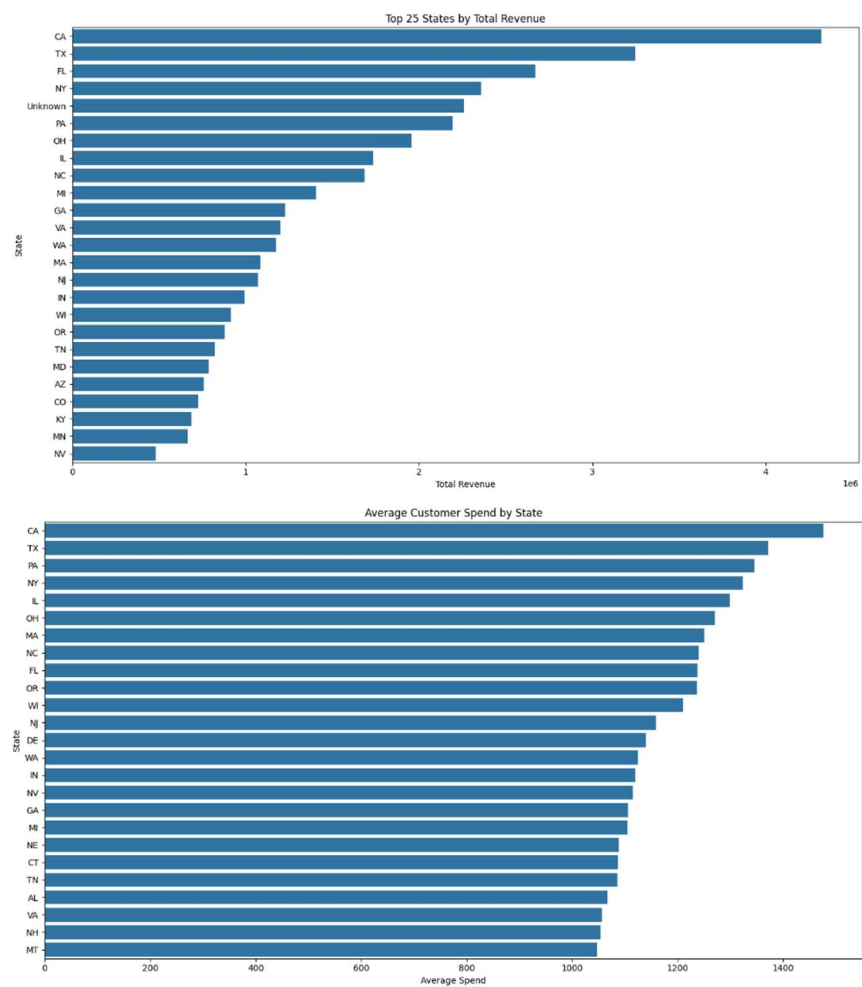


Fig 12 — Top 25 states by total revenue and top 20 by avg spend per customer

- CA, TX, NY collectively account for ~33% of total revenue — driven by population scale.
- States like DC, CT, and MA lead in average spend per customer, reflecting higher-income demographics.
- GA, CO, and MI represent high-growth opportunity markets with a sizable base and room to expand.

5. Customer Segmentation — RFM Analysis & K-Means Clustering

5.1 RFM Metric Computation

RFM was computed at the Survey ResponseID level using the latest order date in the dataset (2024-08-15) as the reference point:

Metric	Formula	Interpretation
Recency (R)	<code>datediff(2024-08-15, max(Order_Date))</code>	Days since last purchase — lower = more recent = better

Metric	Formula	Interpretation
Frequency (F)	count(Order_Date)	Number of purchases — higher = more loyal
Monetary (M)	sum(Revenue)	Total spend — higher = more valuable

5.2 Pre processing

- log1p() transformation applied to all three metrics to reduce right-skew before clustering.
- StandardScaler (z-score normalisation) applied to ensure equal feature weighting in KMeans.
- 16,072 unique customers entered the RFM model.

5.3 Elbow Method

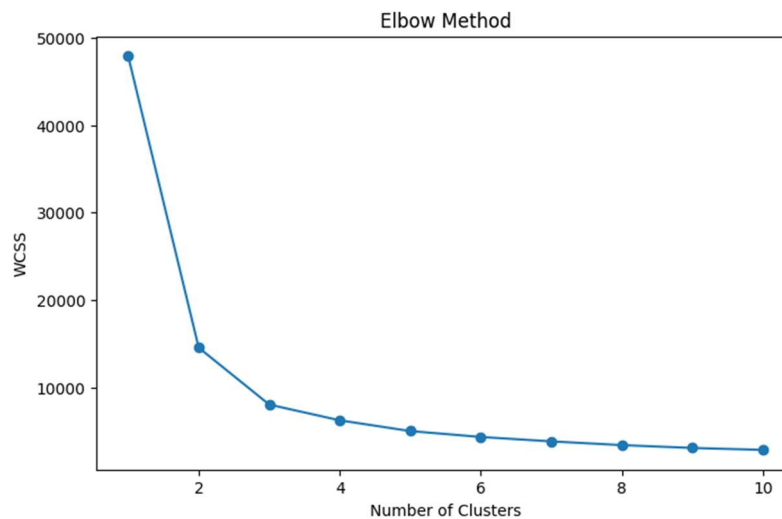


Fig 13 — Elbow method: WCSS vs number of clusters

The elbow is clearly visible at k = 4, confirming four meaningful customer segments.

5.4 Cluster Profiles

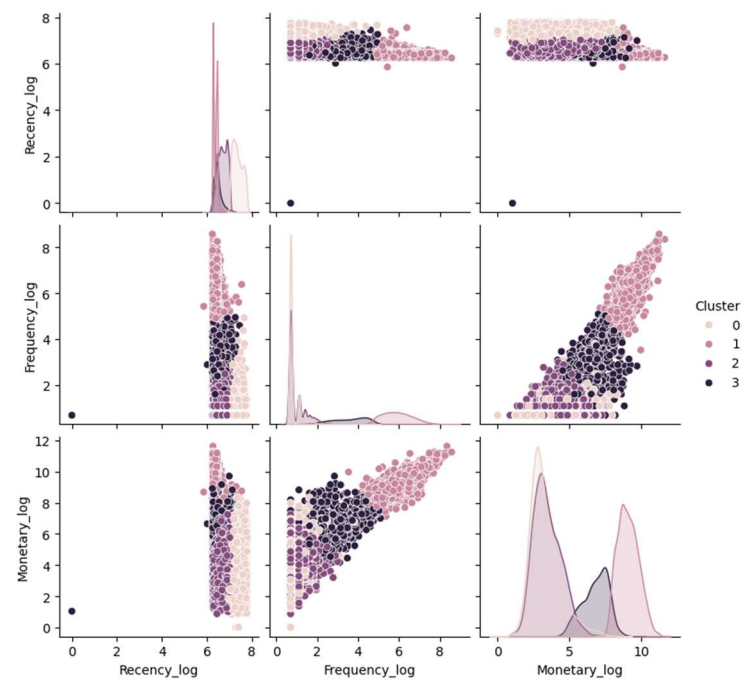


Fig 14 — Customer segments: Frequency vs Recency (log scale)

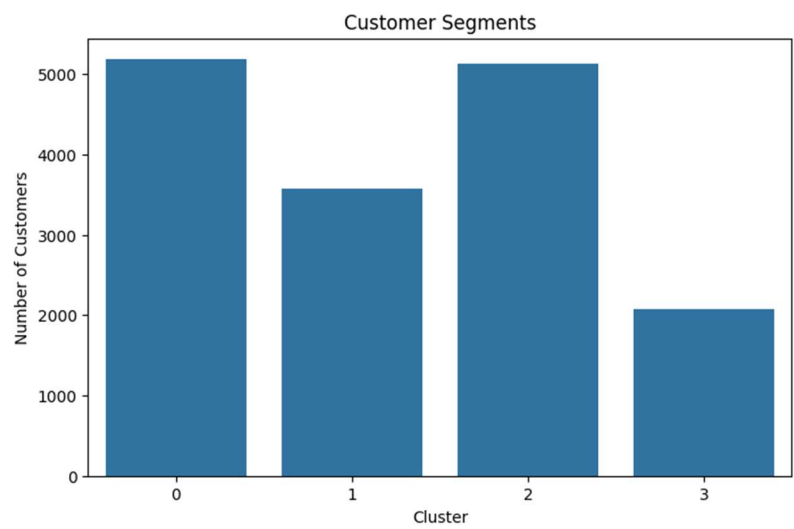


Fig 15 — Average total spend by segment

Segment	Recency	Frequency	Monetary	Strategy
Champions	Low (recent)	Very High	Very High	Reward with loyalty perks; early access; VIP tiers
Potential Loyalists	Low–Moderate	Moderate	Moderate	Personalised recommendations; subscription upsell
At-Risk / Dormant	High (lapsed)	Low	Low–Moderate	Win-back campaigns with urgency; limited-time offers
Low-Value / New	Moderate	Very Low	Very Low	Onboarding campaigns; first-purchase incentives

5.5 Cluster Distribution by Income

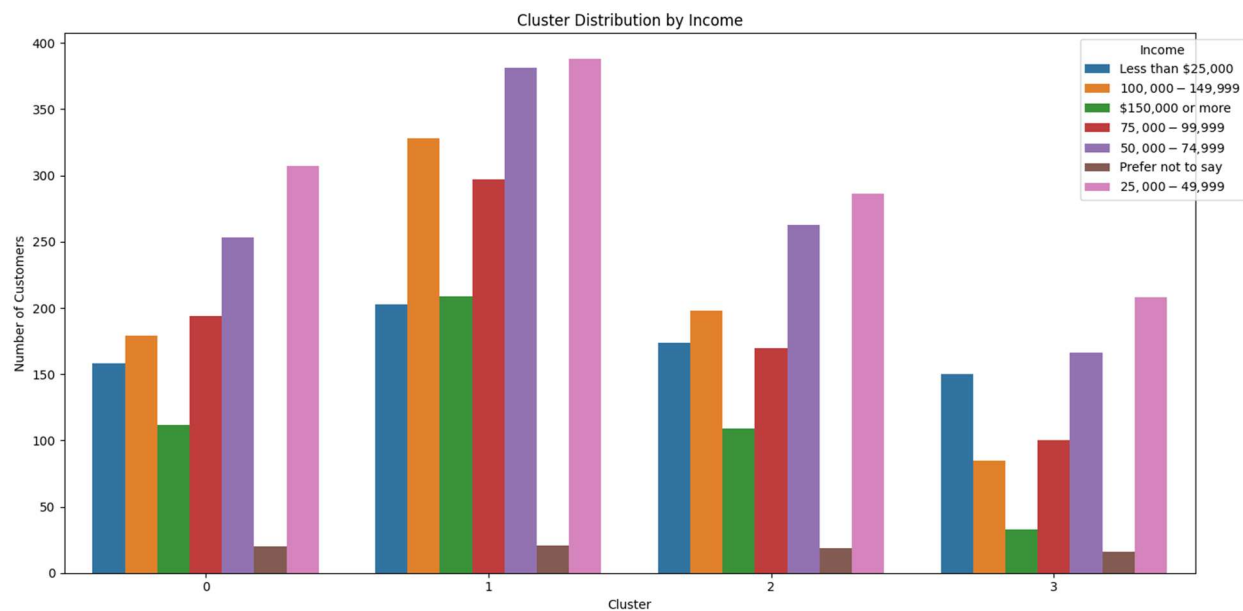


Fig 16 — Income bracket distribution within each customer segment

- Champions and Potential Loyalists skew toward mid-to-high income (\$75K–\$150K+).
- At-Risk customers are distributed more evenly across income levels — lapse is behaviour-driven, not purely financial.

5.6 Top Categories by Segment

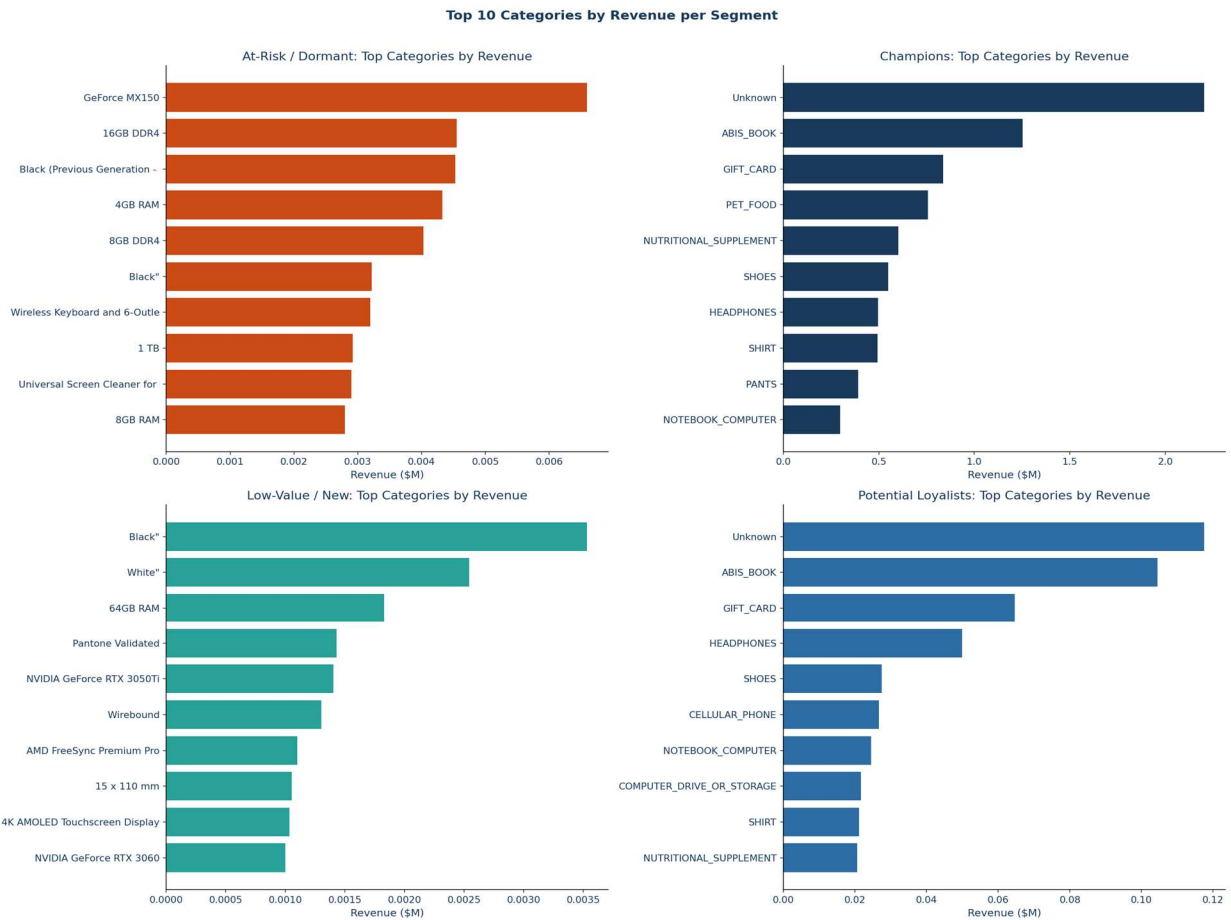


Fig 17 — Top 10 revenue-generating categories per customer segment

- **Champions:** Health & Beauty, Electronics, and Office Products — repeat, high-value categories.
- **At-Risk:** concentrated in Books and Grocery — lower-commitment, lower-price categories.
- **Potential Loyalists:** diverse category mix indicating exploration phase.
- **Low-Value / New:** narrow category exposure, often a single first-purchase category.

6. Conclusion & Recommendations

Through data preparation, cleaning, EDA, and customer segmentation using PySpark and K-Means clustering, this analysis produced the following strategic recommendations:

Focus Area	Core Finding	Action
Inventory	Nov–Dec = 30–35% of annual revenue; Office Products spike Aug–Sep; Grocery is flat year-round	Pre-position stock October for Q4; replenish Office Products July; steady Grocery supply chain

Focus Area	Core Finding	Action
Retention	95%+ repeat buyers; 4-cluster RFM identifies Champions, At-Risk, Potential Loyalists, Low-Value	Cluster-specific campaigns: VIP rewards for Champions; win-back for At-Risk; nurture for Loyalists
Pricing	High-income customers cluster in Champion and Loyalist segments; Low-Value concentrates in consumables	Premium subscription tiers for Champions; first-purchase incentives for Low-Value to move up funnel
Geography	CA+TX+NY $\approx$ 33% revenue; DC/CT/MA lead per-customer spend; GA/CO/MI = growth markets	35–40% ad budget in top-3 states; premium campaigns in high-spend states; growth push in mid-tier

The integration of Apache Spark for distributed data processing, PySpark DataFrame operations for transformation and aggregation, and scikit-learn KMeans for RFM-based customer segmentation enabled a robust, end-to-end analytical pipeline on a dataset of 1.85 million transactions. E-commerce businesses that act on these findings stand to improve retention, reduce inventory costs, and maximise revenue per customer through data-driven decision-making.

Appendix: Assumptions & Limitations

Item	Detail
Dataset Scope	5,027 US Amazon users (2018–2022). Generalisability to other markets or platforms is limited.
Missing Survey Data	Many purchase records lack matching survey responses. Demographic analyses cover only the subset with survey data.
Revenue Proxy	Revenue = Price $\times$ Quantity. Returns, discounts, and taxes are not accounted for.
Fraud Threshold	Mean + 3σ is a statistical heuristic, not a domain-calibrated rule. False positives expected.
RFM Reference Date	2024-08-15 (latest order date in dataset). In production, today's date would be used.
Bulk Purchase Definition	Quantity > 5 per line item. This threshold is adjustable based on business context.
Clustering Assumption	KMeans assumes spherical clusters. Log-transform + standardisation mitigates but doesn't eliminate this limitation.
Market Basket Sample	5% random sample (seed=42) used for pair analysis due to combinatorial scale of 1.85M rows.

